6 A solenoid is connected to an e.m.f source and a variable resistor as shown in Figure 6.1. A small coil connected to a galvanometer is placed close to the solenoid.

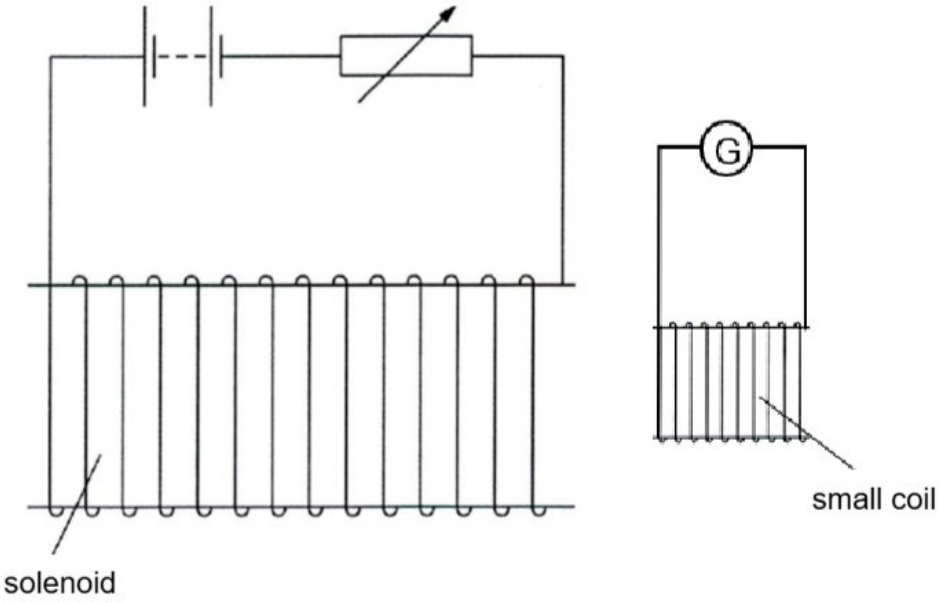


Figure 6.1

(a) The resistance of the variable resistor is slowly increased.

(i) State Faraday's law of electromagnetic induction and use it to explain why a reading is detected in the galvanometer.

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.....

..... [2]

(ii) On Figure 6.1, mark the direction of the induced current passing through the small coil.

Explain the direction of the induced current.

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.....

..... [2]

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(b) The current in the solenoid is now made to vary as shown in Figure 6.2.

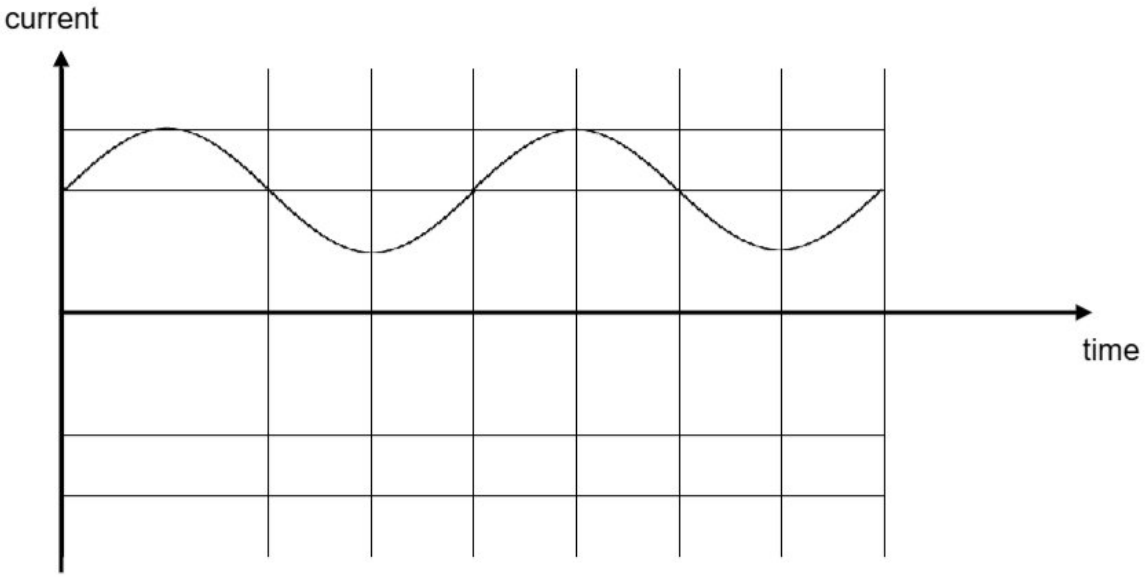


Figure 6.2

(i) On Figure 6.3, sketch the variation with time of the e.m.f. induced in the small coil. Label it as **P**. The time scale is the same. [1]

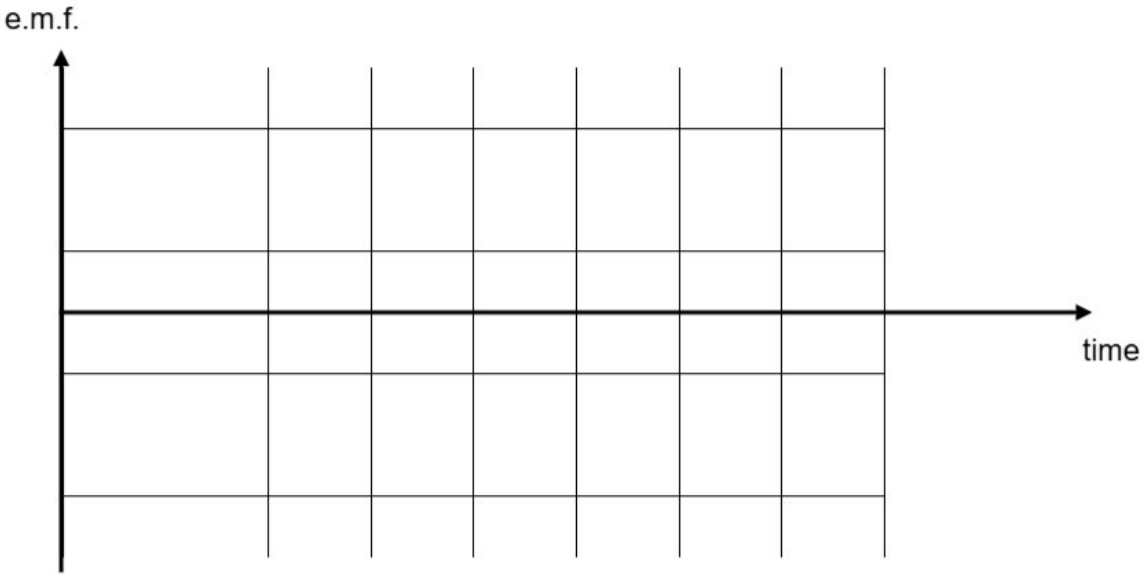


Figure 6.3

(ii) The frequency of the current in the solenoid is halved.

On Figure 6.3, sketch the new variation of the induced e.m.f. in the small coil and label it as **Q**.

[2]

[Total: 7]